TWO INDECOMPOSABILITY THEOREMS FOR 0-HECKE MODULES: INDUCED SIMPLES (Q72(I)) AND KIM–LEE–OH CONJECTURE 7.5 (Q73)

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Summary. Two results on induction products of 0-Hecke modules, proved by one skeleton (Frobenius reciprocity for End and for the top, the Mackey double-coset splitting, and an induction on size).

- **Theorem A (Q72(i)).** $\mathrm{Ind}_{H_a(0)\otimes H_b(0)}^{H_{a+b}(0)}(C_I \boxtimes C_J)$ is indecomposable unless $I, J \in \{(m), (1^m)\}$, in which case it is $P_{I.J} \oplus P_{I\odot J}$.
- **Theorem B (Q73 = Kim–Lee–Oh, arXiv:2310.20571, Conj. 7.5).** If a skew shape is disconnected and no two consecutive connected components are ribbons, then $X_{\lambda/\mu}$ is indecomposable. Strong form: Corollary C.

Grades: both proved (internal grade); each has passed one internal cold read (Dream 288 for Q72(i), Dream 289 for Q73), and the nits found there are fixed in this write-up. Neither has had an external check. Q73 relies on cited results of KLO, listed in §15.
What I want from you: a referee-style check of both proofs, especially **Lemma II.5** (Key Lemma and the return map ζ; Lemma 3 in the PROVE 290 record) and **Lemma II.8** (support lemma; Lemma 5 in the record) of Part II.

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1. Conventions and the shared word model

Throughout, $\Bbbk$ is a field. $H_N = H_N(0)$ is the $\Bbbk$-algebra generated by $\pi_1, \dots, \pi_{N-1}$ with $\pi_i^2 = \pi_i$ and the braid relations. For $w \in S_N$ put $\pi_w := \pi_{i_1} \cdots \pi_{i_\ell}$ for any reduced word $s_{i_1} \cdots s_{i_\ell}$ of w. The basic multiplication rule is

$$\pi_i \pi_w = \begin{cases} \pi_{s_i w}, & \ell(s_i w) > \ell(w), \\ \pi_w, & \ell(s_i w) < \ell(w). \end{cases} \tag{1}$$

For integers $p \le q$ we write $W_{[p,q]} := \langle s_p, \dots, s_q \rangle$. This is a *generator interval*: $W_{[p,q]}$ is the symmetric group on the values $\{p, \dots, q+1\}$ and fixes all other values. For $Z \subseteq [N-1]$, W_Z is the parabolic subgroup generated by $\{s_z : z \in Z\}$.

1.1. **Simples, compositions, projectives.** For $D \subseteq [N-1]$ let C_D be the one-dimensional module on which π_i acts by $[i \in D]$. If $w \in S_N$, then π_w acts on C_Z by $[w \in W_Z]$, because the letters of any reduced word of w form its support. A composition $I \vDash m$ is identified with its descent (partial-sum) set $D_I \subseteq [m-1]$, and we write $C_I := C_{D_I}$. For $I \vDash a$ and $J \vDash b$, the concatenation $I \cdot J$ and near-concatenation $I \odot J$ have

$$D_{I \cdot J} = D_I \cup \{a\} \cup (a + D_J), \qquad D_{I \odot J} = D_I \cup (a + D_J).$$

P_D (or P_I) denotes the projective cover of C_D.

Fact 1.1 (Norton 1979)**.** *The C_D, $D \subseteq [m-1]$, are all the simple H_m-modules, and* $\dim P_D = \#\{w \in S_m : \mathrm{Des}(w) = D\}$. *Hence C_D is projective iff $D \in \{\emptyset, [m-1]\}$, i.e. iff $I \in \{(m), (1^m)\}$. "Projective" for a composition always means this.*

1.2. **The word model of an induced module.** Fix $a, b \ge 1$, $N = a+b$, and let $B := H_a \otimes H_b \subseteq H_N$ be the subalgebra generated by the π_i with $i \ne a$; here π_{a+q} is the q-th generator of H_b. For a B-module V put $\mathrm{Ind}\, V := H_N \otimes_B V$, and for an H_a-module X and an H_b-module Y put $X \boxtimes Y := \mathrm{Ind}(X \otimes Y)$.

Let $W^{(a,b)}$ be the set of *shuffles*: the $w \in S_N$ that are increasing on $[1, a]$ and on $[a+1, N]$. These are the minimal-length left coset representatives of $S_a \times S_b$. H_N is free as a right B-module on $\{T_w := \pi_w : w \in W^{(a,b)}\}$, so $\mathrm{Ind}\, V = \bigoplus_{w \in W^{(a,b)}} T_w \otimes V$.

We encode a shuffle w by a word $x(w) = x_1 \cdots x_N \in \{A, B\}^N$ indexed by *values*: $x_v = A$ iff $w^{-1}(v) \le a$. Every word with exactly a letters A comes from exactly one shuffle.

Lemma 1.2 (action rules)**.** *Let $w \in W^{(a,b)}$, $v \in V$, and $1 \le i \le N-1$. According to the letters (x_i, x_{i+1}) of $x(w)$:*

(i) *AB: $\pi_i(T_w \otimes v) = T_{s_i w} \otimes v$, and the word of $s_i w$ has BA in positions $i, i+1$;*
(ii) *BA: $\pi_i(T_w \otimes v) = T_w \otimes v$;*
(iii) *AA: $\pi_i(T_w \otimes v) = T_w \otimes \pi_p v$, where p is the rank of i among the A-values;*
(iv) *BB: $\pi_i(T_w \otimes v) = T_w \otimes \pi_{a+q} v$, where q is the rank of i among the B-values.*

Proof. (i) Here $w^{-1}(i) \le a < w^{-1}(i+1)$, so $\ell(s_i w) = \ell(w) + 1$. The permutation $s_i w$ exchanges two values lying in different blocks, so it is still increasing on both blocks, i.e. a shuffle. By (1), $\pi_i T_w = T_{s_i w}$. (ii) Here $w^{-1}(i) > w^{-1}(i+1)$, so s_i is a left descent of w, and $\pi_i T_w = T_w$ by

(1). (iii) The values $i, i+1$ are consecutive A-values. Since w is increasing on $[1,a]$, they occupy consecutive positions $p, p+1$, where p is the rank of i among the A-values. Then $s_i w = w s_p$ with $\ell(ws_p) = \ell(w)+1$ (this is Deodhar's lemma in this case), so $\pi_i T_w = T_{ws_p} = T_w \pi_p$. Since $p \le a-1$, $\pi_p \in B$. (iv) is the same as (iii), with positions $a+q, a+q+1$. □

Lemma 1.3 (Mackey splitting)**.** *For a shuffle w let $k(w) := \#\{v \le a : x_v = B\}$, so $0 \le k(w) \le \min(a,b)$. Put $M_k := \bigoplus_{k(w)=k} T_w \otimes V$. Then $\mathrm{Res}_B \mathrm{Ind}\, V = \bigoplus_k M_k$ as B-modules, and $M_0 = T_{\mathrm{id}} \otimes V \cong V$.*

Proof. For $i \ne a$, each rule of Lemma 1.2 either keeps the word or exchanges x_i and x_{i+1}, and $i, i+1$ lie on the same side of the cut between a and $a+1$. So $k(w)$ is preserved. The only word with $k=0$ is $A^a B^b$, which is the word of $w = \mathrm{id}$. □

Remark 1.4. Lemma 1.3 is the standard double-coset (Mackey) decomposition of $\mathrm{Res}_B \mathrm{Ind}_B$: the M_k are indexed by the $(S_a \times S_b)$-double cosets. It is a direct sum for the Hecke algebra at every q, not just at $q=0$. What $q=0$ adds is the *shape* of the pieces, recorded in Lemma 1.5: every basis vector is sent by each π_i to another basis vector or to a scalar multiple of an element of V in the same slot, and this makes M_k a tensor product of two smaller induced modules of the same type.

Lemma 1.5 (cutting a Mackey block)**.** *Let $k \ge 1$. Cut each word as $x = (\ell \mid r)$ with $\ell = x_1 \cdots x_a$ (containing $a-k$ letters A and k letters B) and $r = x_{a+1} \cdots x_N$ (containing k letters A and $b-k$ letters B). Every pair (ℓ, r) of such words occurs. On $T_{(\ell|r)} \otimes v$:*

(a) *for $i < a$, π_i changes only ℓ. In case AA it acts on v by π_p, where $p \le a-k-1$ is the rank of i among the A's of ℓ. In case BB it acts by π_{a+q}, where $q \le k-1$ is the rank of i among the B's of ℓ;*

(b) *for $i = a+t$ with $1 \le t \le b-1$, π_i changes only r. In case AA it acts by $\pi_{(a-k)+p'}$, where $p' \in [1, k-1]$ is the rank of t among the A's of r. In case BB it acts by $\pi_{a+k+q'}$, where $q' \in [1, b-k-1]$ is the rank of t among the B's of r.*

Proof. Any word with a letters A is a shuffle word, so every pair occurs. Every value in $[1,a]$ is below every value in $[a+1, N]$. Hence the rank of a left letter among all A-values (resp. B-values) equals its rank inside ℓ. A right A-letter has overall A-rank $(a-k)+$(its rank in r), and a right B-letter has overall B-rank $k+$(its rank in r). Now apply Lemma 1.2. The bounds hold because both letters i and $i+1$ are counted. □

Part I. Induced simples: Q72(i)

2. Statement

Fix $I \vDash a$ and $J \vDash b$ with $a, b \ge 1$, $N = a+b$. Let $V = C_I \boxtimes C_J = \Bbbk v$ as a B-module, so π_i $(i \ne a)$ acts on v by $\chi(i) := [i \in D]$, where

$$D := D_I \cup (a + D_J) \subseteq [N-1] \setminus \{a\}.$$

Put $M = M_{I,J} := H_N \otimes_B V$ and $g := 1 \otimes v$. By §1, M has basis $e_w := T_w \otimes v$ $(w \in W^{(a,b)})$. In Lemma 1.2(iii),(iv) the actions become scalars: $\pi_i e_w = \chi(p) e_w$ in case AA and $\pi_i e_w = \chi(a+q) e_w$ in case BB.

Theorem A (Q72(i))**.** *$M_{I,J}$ is indecomposable unless I and J are both projective, i.e. $I, J \in \{(m), (1^m)\}$. In the both-projective case $M_{I,J} \cong P_{I \cdot J} \oplus P_{I \odot J}$, with two nonzero, non-isomorphic summands.*

3. End M and the locality criterion

Lemma 3.1 (Frobenius)**.** *$\varphi \mapsto \varphi(g)$ is a linear isomorphism $\operatorname{End}_{H_N}(M) \to E := \{u \in M : \pi_i u = \chi(i)u \text{ for all } i \neq a\}$. Write φ_u for the endomorphism with $\varphi_u(g) = u$.*

Proof. $\operatorname{End}_{H_N}(H_N \otimes_B V, M) \cong \operatorname{Hom}_B(V, \operatorname{Res} M)$, and since $V = \Bbbk v$ with $\pi_i v = \chi(i)v$, a B-map $V \to M$ is the same thing as a choice of image $u \in E$. □

Lemma 3.2 (top)**.** $\operatorname{Hom}_{H_N}(M, C_Z) \cong \operatorname{Hom}_B(V, \operatorname{Res} C_Z)$. *This is one-dimensional if $Z \setminus \{a\} = D$ and zero otherwise. Hence* $\operatorname{top} M = C_{Z_1} \oplus C_{Z_2}$ *with $Z_1 = D = D_{I \odot J}$ and $Z_2 = D \cup \{a\} = D_{I.J}$, and these are distinct since $a \notin D$. Let $f_j : M \to C_{Z_j}$ be the unique module maps with $f_j(g) = 1$. Then*

$$f_j(e_w) = [w \in W_{Z_j}].$$

Proof. The first isomorphism is Frobenius reciprocity. $\operatorname{Res} C_Z$ is the character $i \mapsto [i \in Z]$ $(i \neq a)$, which equals χ iff $Z \setminus \{a\} = D$. Simples are one-dimensional with End $= \Bbbk$, so the multiplicity of C_Z in $\operatorname{top} M$ is $\dim \operatorname{Hom}(M, C_Z)$. Finally, $f_j(e_w) = f_j(\pi_w g) = \pi_w \cdot 1 = [w \in W_{Z_j}]$ in C_{Z_j}. □

Proposition 3.3 ((Crit))**.** *M is indecomposable if and only if $f_1(u) = f_2(u)$ for every $u \in E$.*

Proof. Each $\varphi \in \operatorname{End} M$ induces an endomorphism of $\operatorname{top} M = C_{Z_1} \oplus C_{Z_2}$. Since the top is multiplicity-free, this endomorphism is $\operatorname{diag}(c_1, c_2)$, where $f_j \circ \varphi = c_j f_j$, so $c_j = f_j(\varphi(g))$. This defines an algebra map $\rho : \operatorname{End} M \to \Bbbk \times \Bbbk$.

The kernel $\ker \rho$ is $\{\varphi : \varphi(M) \subseteq \operatorname{rad} M\}$. Every such φ is nilpotent, because $\varphi(\operatorname{rad}^j M) = \operatorname{rad}^j \varphi(M) \subseteq \operatorname{rad}^{j+1} M$. Hence $\ker \rho$ is a nil ideal, and $\ker \rho \subseteq J(\operatorname{End} M)$.

The image $\operatorname{im} \rho$ is a subalgebra of $\Bbbk \times \Bbbk$ containing $(1,1)$, so it is either $\Bbbk(1,1)$ or $\Bbbk \times \Bbbk$.

- If $\operatorname{im} \rho = \Bbbk \times \Bbbk$, the idempotent $(1,0)$ lifts modulo the nil ideal $\ker \rho$ to a nontrivial idempotent of $\operatorname{End} M$, so M is decomposable.
- If $\operatorname{im} \rho = \Bbbk(1,1)$, then $\operatorname{End} M / \ker \rho \cong \Bbbk$ with $\ker \rho$ nil, so $\operatorname{End} M$ is local and M is indecomposable.

Finally, $\operatorname{im} \rho = \Bbbk(1,1)$ iff $c_1 = c_2$ for all φ. By Lemma 3.1 this means $f_1(u) = f_2(u)$ for all $u \in E$. □

4. Computing f_1 and f_2

Lemma 4.1. (a) *$f_1(e_w) = [w = \mathrm{id}]$, so $f_1(u) = u_{\mathrm{id}}$ (the coefficient of $e_{\mathrm{id}} = g$).*

(b) *Let $[\alpha, \beta] \ni a$ be the component of Z_2 containing a (as a generator interval). Then $\alpha = a - r'$ and $\beta = a + s'$, where $r' :=$ the length of the run $a-1, a-2, \ldots$ in D_I and $s' :=$ the length of the run $1, 2, \ldots$ in D_J. For a shuffle w,*

$$f_2(e_w) = [x_1 = \cdots = x_{\alpha-1} = A] \cdot [x_{\beta+2} = \cdots = x_N = B]. \tag{2}$$

Proof. (a) Since $a \notin Z_1$, we have $W_{Z_1} \subseteq S_a \times S_b$, and the only shuffle in $S_a \times S_b$ is id.

(b) The description of $[\alpha, \beta]$ is immediate from $Z_2 = D_I \cup \{a\} \cup (a + D_J)$. Write W_{Z_2} as the commuting product of the W_K over the components K of Z_2, and write $w \in W_{Z_2}$ as $w = \prod_K w_K$. The right descent set of w is the union of those of the w_K, and $w_K \neq e$ has a right descent in K. A shuffle has right descents only at a. So every factor w_K with $a \notin K$ is trivial, and a shuffle lies in W_{Z_2} iff it lies in $W_{[\alpha,\beta]}$, i.e. iff it fixes every value $< \alpha$ and every value $> \beta + 1$.

Now let w be a shuffle. If w fixes $1, \ldots, \alpha - 1$, then these values sit in positions $\leq \alpha - 1 \leq a$, so they are A-values. Conversely, if $1, \ldots, \alpha - 1$ are A-values, then, since w is increasing on $[1, a]$, they are the $\alpha - 1$ smallest A-values and sit in positions $1, \ldots, \alpha - 1$. Symmetrically, using $\beta + 2 \geq a + 2$, w fixes every value $> \beta + 1$ iff all those values are B-values. □

5. Mackey factorisation of E

By Lemma 1.3, $\mathrm{Res}_B M = \bigoplus_k M_k$ with $M_0 = \Bbbk g$. Since the π_i $(i \neq a)$ preserve each M_k, we get $E = \bigoplus_k (E \cap M_k)$.

Lemma 5.1. *For $k \geq 1$, the map $e_{(\ell|r)} \mapsto e_\ell \otimes e_r$ is a B-module isomorphism $M_k \cong L_k \otimes R_k$, where*

$$L_k := \mathrm{Ind}_{H_{a-k} \otimes H_k}^{H_a}\big(C_{I^{(1)}} \boxtimes C_{J^{(1)}}\big), \qquad R_k := \mathrm{Ind}_{H_k \otimes H_{b-k}}^{H_b}\big(C_{I^{(2)}} \boxtimes C_{J^{(2)}}\big),$$
$$D_{I^{(1)}} = D_I \cap [1, a-k-1], \qquad D_{I^{(2)}} = \{p - (a-k) : p \in D_I,\ p > a-k\},$$
$$D_{J^{(1)}} = D_J \cap [1, k-1], \qquad D_{J^{(2)}} = \{q - k : q \in D_J,\ q > k\}.$$

Degenerate cases: $L_a = C_{J^{(1)}}$ (word B^a) and $R_b = C_{I^{(2)}}$ (word A^b).

Proof. Compare Lemma 1.5 with the word models of L_k and R_k. On the left, AA at $i < a$ gives the eigenvalue $\chi(p) = [p \in D_I]$ with $p \leq a-k-1$, i.e. $[p \in D_{I^{(1)}}]$. BB gives $\chi(a+q) = [q \in D_J]$ with $q \leq k-1$, i.e. $[q \in D_{J^{(1)}}]$. On the right, AA gives $[(a-k)+p' \in D_I] = [p' \in D_{I^{(2)}}]$, and BB gives $[k+q' \in D_J] = [q' \in D_{J^{(2)}}]$. The AB/BA rules agree verbatim. H_a acts only on ℓ and H_b only on r, and the basis of M_k is the product of the two sets of words. □

Eigenspaces multiply. Let $U_k := \{x \in L_k : \pi_i x = [i \in D_I]x \ \forall i < a\} \cong \mathrm{Hom}(C_I, L_k)$ and $U'_k := \{y \in R_k : \pi_t y = [t \in D_J]y \ \forall t < b\} \cong \mathrm{Hom}(C_J, R_k)$. Then

$$E \cap M_k = U_k \otimes U'_k.$$

Indeed, $\bigcap_i \ker((A_i - c_i) \otimes 1) = (\bigcap_i \ker(A_i - c_i)) \otimes R_k$ for commuting operators A_i on L_k, likewise on the right, and $(U \otimes R_k) \cap (L_k \otimes U') = U \otimes U'$.

f_2 on a block. By (2), and since $\alpha - 1 \leq a - 1$ and $\beta + 2 \geq a + 2$, we get $f_2|_{M_k} = \lambda_k \otimes \lambda'_k$, where

$$\lambda_k(e_\ell) = [\ell_1 = \cdots = \ell_{\alpha-1} = A], \qquad \lambda'_k(e_r) = [r_{\beta+2-a} = \cdots = r_b = B].$$

f_2 is an H_N-map to C_{Z_2}, and $\mathrm{Res}_B C_{Z_2} = C_I \boxtimes C_J$. So $f_2|_{M_k}$ is a B-map $L_k \otimes R_k \to C_I \otimes C_J$. If $\lambda'_k \neq 0$, choose y with $\lambda'_k(y) = 1$. Then $\lambda_k(\pi_i x) = (\lambda_k \otimes \lambda'_k)(\pi_i x \otimes y) = [i \in D_I]\lambda_k(x)$, so $\lambda_k : L_k \to C_I$ is a module map. Symmetrically, if $\lambda_k \neq 0$ then $\lambda'_k : R_k \to C_J$ is a module map.

Proposition 5.2. *If M is decomposable, then there is $k \geq 1$ with $\lambda_k(U_k) \neq 0$ and $\lambda'_k(U'_k) \neq 0$. For such k:*

(a) *C_I is a direct summand of L_k, and C_J is a direct summand of R_k;*
(b) *$k \leq r' + 1$.*

Proof. On $M_0 = \Bbbk g$ we have $f_1 = f_2$, and for $k \geq 1$, f_1 vanishes on M_k by Lemma 4.1(a). So for $u = \sum_k u_k \in E$ with $u_k \in E \cap M_k$, $f_2(u) - f_1(u) = \sum_{k \geq 1} f_2(u_k)$. If M is decomposable, Proposition 3.3 gives $u \in E$ with $f_1(u) \neq f_2(u)$, hence some $k \geq 1$ with $(\lambda_k \otimes \lambda'_k)(U_k \otimes U'_k) \neq 0$.

(a) Both λ_k and λ'_k are nonzero, so both are module maps. An $x \in U_k$ with $\lambda_k(x) \neq 0$ spans a copy of C_I, and the composite $C_I \to L_k \xrightarrow{\lambda_k} C_I$ is nonzero, hence an isomorphism. So C_I splits off L_k. The same argument works for C_J and R_k.

(b) $\lambda_k(e_\ell) \neq 0$ requires all k letters B of ℓ to lie in positions $[\alpha, a]$, which has $a - \alpha + 1 = r' + 1$ slots. □

6. Induction on N and the converse

We induct on N. The base case is $N = 2$, i.e. $a = b = 1$, where both simples are projective. Assume Theorem A holds for all sizes $< N$, suppose $M_{I,J}$ is decomposable, and take k as in Proposition 5.2.

Lemma 6.1. *If $1 \le k \le a-1$ and C_I is a summand of L_k, then I is projective. Likewise, if $1 \le k \le b-1$ and C_J is a summand of R_k, then J is projective.*

Proof. $L_k = M_{I^{(1)},J^{(1)}}$ is a module of size $a < N$, so the induction hypothesis applies. If L_k is indecomposable, then $L_k \cong C_I$, so $\dim L_k = \binom{a}{k} = 1$. That is impossible for $1 \le k \le a-1$. Otherwise L_k is decomposable, so by induction $I^{(1)}, J^{(1)}$ are projective and $L_k \cong P \oplus P'$ is projective. Then C_I, as a summand of a projective, is projective. The right-hand statement has the same proof. □

Recall $k \le \min(a,b)$. We split into cases.

- $k < a$ and $k < b$. Lemma 6.1 applies on both sides, so I and J are projective.
- $k = a < b$. J is projective by Lemma 6.1. On the left, $C_I \mid L_a = C_{J^{(1)}}$ gives $D_I = D_J \cap [1, a-1]$. Since $D_J \in \{\emptyset, [b-1]\}$, this is $\emptyset$ or $[a-1]$, so I is projective.
- $k = b < a$. Symmetrically, I is projective and $D_J = D_{I^{(2)}} \in \{\emptyset, [b-1]\}$.
- $k = a = b$. Here $L_a = C_{J^{(1)}}$ with $D_{J^{(1)}} = D_J$, so $D_I = D_J$. Also, Proposition 5.2(b) gives $a \le r' + 1$, so $r' = a-1$ and $[1, a-1] \subseteq D_I$. Hence $I = J = (1^a)$, which is projective.

So decomposability forces both I and J to be projective.

Converse. If I and J are projective, then $C_I = P_I$ and $C_J = P_J$. $P_I \otimes P_J$ is a projective B-module, and H_N is free over B, so $M = \mathrm{Ind}(P_I \otimes P_J)$ is projective. The multiplicity of P_Z in M is $\dim \mathrm{Hom}(M, C_Z)$, which by Lemma 3.2 is 1 exactly for $Z \in \{Z_1, Z_2\}$. So $M \cong P_{I \cdot J} \oplus P_{I \odot J}$. □

7. Remarks on Part I

Remark 7.1 (where $q = 0$ is used)**.** The proof applies Frobenius reciprocity twice: once to $\mathrm{End}\, M$ and once to $\mathrm{top}\, M$. The Mackey decomposition $\mathrm{Res}_B M = \bigoplus_k M_k$ is the double-coset decomposition. It is a direct sum at *every* q (Remark 1.4), so its directness is not where $q = 0$ enters. The proof uses $q = 0$ in three places:

- the specific shape of the pieces: $M_k \cong L_k \otimes R_k$ with explicitly matched eigenvalues (Lemma 5.1), which rests on the 0-Hecke rules of Lemma 1.2;
- the fact that the simples are one-dimensional characters C_D;
- Norton's projectivity criterion (Fact 1.1).

Mackey then reduces the gluing question to "does C_I split off a smaller induced module L_k?", and induction answers it.

Remark 7.2. The bound $k \le r' + 1$ is used only in the case $k = a = b$. It is what excludes, for example, $I = J = (2,2)$, where $\dim \mathrm{End}\, M = 4$ but M is indecomposable.

Remark 7.3 (negative control)**.** In the both-projective case a witness k does exist. For example, take $k = 1$ with $a, b \ge 2$: then L_1 and R_1 are projective and contain C_I and C_J. In the non-projective case, the hypothesis that fails is "C_I is a summand of L_k". The Wake 288 guess ("I has both an ascent and a descent") is the $k < a$ shadow of this condition, via Fact 1.1.

Remark 7.4. Hypotheses used: $q = 0$ (as in Remark 7.1), the two-block parabolic, and one-dimensional factors. Q72(ii), which asks for a description of $\mathrm{End}\, M$ in general, is untouched. The proof does describe $\mathrm{End}\, M$ modulo its radical: $\mathrm{End}\, M = \Bbbk \oplus \ker \rho$.

Part II. Kim–Lee–Oh Conjecture 7.5: Q73

8. Statement

KLO is Kim–Lee–Oh, *Regular Schur labeled skew shape posets and their* 0*-Hecke modules*, arXiv:2310.20571. For a basic skew shape λ/μ with n cells, $X_{\lambda/\mu}$ is the H_n-module with basis

$\mathrm{SYT}(\lambda/\mu)$ (KLO App. A). For an SYT T and $1 \le i \le n-1$:

$$\pi_i T = \begin{cases} T, & i \text{ strictly left of } i+1, \\ 0, & i \text{ and } i+1 \text{ in the same column}, \\ s_i T, & i \text{ strictly right of } i+1. \end{cases}$$

KLO show that $X_{\lambda/\mu} \cong M_P$ whenever $\mathrm{sh}(\tau_P) = \lambda/\mu$. Write $\lambda/\mu = C_1 \star \cdots \star C_m$ for its connected components, with C_1 the north-east one. The components occupy pairwise disjoint row ranges and column ranges, and C_s lies strictly north-east of C_t for $s < t$. A component is a *ribbon* if it contains no 2×2 square.

Theorem B (KLO Conjecture 7.5)**.** *If $m \ge 2$ and no two consecutive components C_s, C_{s+1} are both ribbons, then $X_{\lambda/\mu}$ is indecomposable.*

Corollary C (strong form)**.** *For an arbitrary skew shape λ/μ, $X_{\lambda/\mu} \cong \bigoplus_S X_S$ with every X_S indecomposable. Here S runs over the shapes obtained by replacing each maximal run of r consecutive ribbon components by one connected ribbon, using one of the 2^{r-1} choices of $\cdot/\odot$ concatenation (KLO Lemma 5.1, Rem. A.1(2)). So the number of indecomposable summands is $\prod 2^{r-1}$.*

9. Word model for shapes, rotation

The notation of §1 is in force, with X an H_a-module, Y an H_b-module and $V = X \otimes Y$. In Lemma 1.2(iv), π_{a+q} acts on Y as π_q.

Lemma II.1 (tableau = word model; KLO Prop. A.2(1), JKLO arXiv:2105.14169 Lemma 4)**.** *Let S, S' be basic skew shapes with $|S| = a$ and $|S'| = b$. The map*

$$T_w \otimes T \otimes T' \;\longmapsto\; \textit{the filling of } S \star S' \textit{ by } w(T) \textit{ on } S \textit{ and } w(a + T') \textit{ on } S'$$

is an isomorphism $X_S \boxtimes X_{S'} \to X_{S \star S'}$.

Proof. Bijection on bases. There are no order relations between S-cells and S'-cells in $S \star S'$. So an SYT of $S \star S'$ is exactly a choice of the value set on S (equivalently a shuffle w) together with two standardised SYT T, T'. The filling is standard because w is increasing on both blocks.

Action. Every S-cell is strictly to the right of every S'-cell, and in a different row. If $i \in S$ and $i+1 \in S'$, the tableau rule gives s_i, which matches word rule AB. If $i \in S'$ and $i+1 \in S$, it gives the identity, matching BA. If $i, i+1 \in S$, they are the images of $p, p+1$ with p the rank of i, and their relative position is that of $p, p+1$ in T. So the tableau rule is the X_S rule for π_p, matching AA. Rule BB is the same on the S' side. □

Lemma II.2 (rotation)**.** *Let T° be T rotated by $180°$, with each value v replaced by $n+1-v$. Then $T \mapsto T^\circ$ gives $X_{(\lambda/\mu)^\circ} \cong \varphi^* X_{\lambda/\mu}$, where φ is the automorphism $\pi_i \mapsto \pi_{n-i}$. Moreover $(C_1 \star C_2)^\circ = C_2^\circ \star C_1^\circ$, and rotation sends ribbons to ribbons. In particular, rotation preserves indecomposability and the hypothesis of Theorem B.*

Proof. Rotation reverses left and right and preserves columns. So "i is strictly left of / in the same column as / strictly right of $i+1$ in T°" is equivalent to "$n-i$ is strictly left of / in the same column as / strictly right of $n-i+1$ in T". □

10. Tops and the indecomposability criterion

Standing hypothesis (H). X and Y are indecomposable, with multiplicity-free tops $\operatorname{top} X = \bigoplus_{i \in \mathcal{I}} C_{D_i}$ and $\operatorname{top} Y = \bigoplus_{j \in \mathcal{J}} C_{E_j}$. Let $\varepsilon_i : X \to C_{D_i}$ and $\delta_j : Y \to C_{E_j}$ be the top functionals, and fix $x_i \in X$ and $y_j \in Y$ with $\varepsilon_i(x_i) = 1$ and $\delta_j(y_j) = 1$.

Lemma II.3 (top)**.** *Assume only that* $\operatorname{top} X$ *and* $\operatorname{top} Y$ *are multiplicity-free. Then* $\operatorname{top}(X \boxtimes Y) = \bigoplus_{i,j,e} C_{Z_{ije}}$ *is multiplicity-free, where* $e \in \{0,1\}$ *and*

$$Z_{ij0} = D_i \cup (a + E_j), \qquad Z_{ij1} = Z_{ij0} \cup \{a\}.$$

The corresponding functional is $f_{ije}(T_w \otimes v) = [w \in W_{Z_{ije}}] \cdot (\varepsilon_i \otimes \delta_j)(v)$.

Proof. Frobenius reciprocity and the tensor identity for finite-dimensional modules give

$$\operatorname{Hom}(X \boxtimes Y, C_Z) \cong \operatorname{Hom}_B(X \otimes Y, \operatorname{Res} C_Z) = \operatorname{Hom}(X, C_{Z \cap [a-1]}) \otimes \operatorname{Hom}(Y, C_{(Z-a)\cap[b-1]}).$$

By multiplicity-freeness this is one-dimensional exactly when $Z = Z_{ije}$ for some i, j, e, and zero otherwise. The extension of $\beta \in \operatorname{Hom}_B(V, \operatorname{Res} C_Z)$ is $T_w \otimes v \mapsto \chi_Z(\pi_w)\beta(v) = [w \in W_Z]\beta(v)$. The sets Z_{ije} are pairwise distinct, since Z determines D_i, E_j and whether $a \in Z$. □

Lemma II.4 (criterion)**.** *Assume (H) and put* $M = X \boxtimes Y$. *Then* M *is decomposable if and only if there exist* $k \geq 1$, $i \in \mathcal{I}$, $j \in \mathcal{J}$ *and* $u \in \operatorname{Hom}_B(V, M_k)$ *with* $f_{ij1}(u(x_i \otimes y_j)) \neq 0$. *Such a tuple* (k, i, j, u) *is called a* witness.

Proof. 1. *The map* ρ. Every $\varphi \in \operatorname{End} M$ acts on the multiplicity-free top by scalars, which gives an algebra map $\rho : \operatorname{End} M \to \Bbbk^{\mathcal{I}\times\mathcal{J}\times\{0,1\}}$. Its kernel consists of maps into $\operatorname{rad} M = \bigcap \ker f_{ije}$, which are nilpotent (as in Proposition 3.3). So $\ker\rho \subseteq J(\operatorname{End} M)$. The image $\operatorname{im}\rho$ is a subalgebra of some $\Bbbk^s$, hence reduced, hence $\cong \Bbbk^t$ and semisimple. Therefore $J(\operatorname{End} M) = \ker\rho$ and $\operatorname{End} M/J \cong \Bbbk^t$, where t is the number of indecomposable summands of M. In particular, M is indecomposable iff $\operatorname{im}\rho = \Bbbk \cdot 1$. The coordinate of $\rho(\varphi)$ at (i, j, e) is $f_{ije}(\varphi(1 \otimes x_i \otimes y_j))$, because $f_{ije}(1 \otimes x_i \otimes y_j) = 1$.

2. *Frobenius.* φ corresponds to $u = \varphi|_{1\otimes V} \in \operatorname{Hom}_B(V, \operatorname{Res} M) = \bigoplus_k \operatorname{Hom}_B(V, M_k)$ (Lemma 1.3). Write $u = \sum_k u_k$. The u_k can be chosen independently.
3. $k = 0$. Here $u_0 \in \operatorname{End}_B(V) = \operatorname{End} X \otimes \operatorname{End} Y$; write $u_0 = \sum_s \alpha_s \otimes \beta_s$. Both rings are local. By step 1 applied to X (respectively Y), every element of $J(\operatorname{End} X)$ is nilpotent, so it lies in $\ker\rho^X$. Hence $\rho^X(\alpha_s) = a_s \cdot 1$ and $\rho^Y(\beta_s) = b_s \cdot 1$ are scalars. Since $f_{ije}|_{M_0} = \varepsilon_i \otimes \delta_j$, each f_{ije} contributes the same scalar $c_0 = \sum_s a_s b_s$ from u_0.
4. $k \geq 1$. f_{ij0} vanishes on M_k: since $a \notin Z_{ij0}$, $W_{Z_{ij0}} \cap W^{(a,b)} = \{\mathrm{id}\}$, while $k \geq 1$ forces $w \neq \mathrm{id}$.
5. *Assembling.* Steps 3 and 4 give $\rho(\varphi)_{ij0} = c_0$ and $\rho(\varphi)_{ij1} = c_0 + \sum_{k\geq 1} f_{ij1}(u_k(x_i \otimes y_j))$. If no witness exists, then $\rho(\varphi) = c_0 \cdot 1$ for every φ. If (k, i, j, u) is a witness, take $u_k = u$ and all other $u_{k'} = 0$. Then $\rho(\varphi)_{ij0} = 0 \neq \rho(\varphi)_{ij1}$, so $\rho(\varphi)$ is not scalar. □

Region formula. Let $[\alpha, \beta]$ be the component of Z_{ij1} containing a, as a generator interval. Then $\alpha = a - r'$ and $\beta = a + s'$, where r' is the length of the run $a-1, a-2, \ldots$ in D_i and s' is the length of the run $1, 2, \ldots$ in E_j. Exactly as in Lemma 4.1(b), for a shuffle w:

$$w \in W_{Z_{ij1}} \iff w \in W_{[\alpha,\beta]} \iff x_1 = \cdots = x_{\alpha-1} = A \ \text{ and } \ x_{\beta+2} = \cdots = x_N = B.$$

11. Key Lemma: a witness makes X a summand of a smaller induced module

Lemma II.5 (Key Lemma)**.** *Assume (H), and let* (k, i, j, u) *be a witness. Then:*

(a) $[a - k + 1, a - 1] \subseteq D_i$ *and* $[1, k - 1] \subseteq E_j$;

(b) X *is a direct summand of*

$$\widehat{L} := \operatorname{Ind}_{H_{a-k}\otimes H_k}^{H_a}\big(\operatorname{Res}_{H_{a-k}} X \boxtimes C_{[k-1]}\big),$$

via H_a*-maps* $X \to \widehat{L} \to X$ *whose composite* g *satisfies* $\varepsilon_i \circ g \neq 0$. *Explicitly* $g = \zeta \circ \Psi''$, *with* ζ *and* Ψ'' *defined in the proof.*

Here $H_{a-k} = \langle \pi_1, \ldots, \pi_{a-k-1} \rangle$ *and* $H_k = \langle \pi_{a-k+1}, \ldots, \pi_{a-1} \rangle$ *inside* H_a.

Proof. Step 1. *M_k as an H_a-module.* By Lemma 1.5(a), the generators π_i with $i < a$ change only the left word ℓ. They act on V by $\pi_p \otimes 1$ with $p \le a-k-1$ (case AA) or by $1 \otimes \pi_q$ with $q \le k-1$ (case BB). So for each right word r, the space $\widetilde{L}_r := \operatorname{span}\{T_{(\ell|r)} \otimes v\}$ is H_a-stable, and $M_k|_{H_a} = \bigoplus_r \widetilde{L}_r$. The word model of §1 identifies every $\widetilde{L}_r$ with

$$\widetilde{L} := \operatorname{Ind}_{H_{a-k}\otimes H_k}^{H_a}\big(\operatorname{Res}_{H_{a-k}} X \boxtimes \operatorname{Res}_{H_k} Y\big), \qquad T_{(\ell|r)} \otimes v \mapsto T_\ell \otimes v,$$

where $\operatorname{Res}_{H_k} Y$ means restriction to $\langle \pi_1, \dots, \pi_{k-1}\rangle \subseteq H_b$.

Step 2. *The functional on a copy.* By the region formula,

$$f_{ij1}(T_{(\ell|r)} \otimes x \otimes y) = [\ell_1 = \cdots = \ell_{\alpha-1} = A] \cdot [r \text{ ok}] \cdot \varepsilon_i(x)\delta_j(y),$$

where "r ok" means $r_{\beta+2-a} = \cdots = r_b = B$. Write "$\ell$ ok" for $\ell_1 = \cdots = \ell_{\alpha-1} = A$. On a copy $\widetilde{L}_r$ with r ok, the functional

$$\widetilde{\lambda}(T_\ell \otimes x \otimes y) := [\ell \text{ ok}]\, \varepsilon_i(x)\delta_j(y)$$

equals $f_{ij1}|_{\widetilde{L}_r}$. Since f_{ij1} is H_N-linear and $\operatorname{Res}_{H_a} C_{Z_{ij1}} = C_{D_i}$, $\widetilde{\lambda}$ is H_a-linear into C_{D_i}. On copies with r not ok, f_{ij1} vanishes.

Step 3. *Choice of copy.* The map $\Psi : X \to M_k$, $x \mapsto u(x \otimes y_j)$, is H_a-linear because u is B-linear. Write $\Psi = \sum_r \Psi_r$ with $\Psi_r : X \to \widetilde{L}_r$. Since $f_{ij1}(\Psi(x_i)) \neq 0$, some copy r (necessarily with r ok) has $\widetilde{\lambda}(\Psi_r(x_i)) \neq 0$. Fix it.

Step 4. *Proof of (a).* For $\widetilde{\lambda} \neq 0$ we need a left word whose k letters B all lie in positions $[\alpha, a]$. So $k \le a - \alpha + 1 = r' + 1$, which gives $[a-k+1, a-1] \subseteq [a-r', a-1] \subseteq D_i$.

For the second inclusion, let $E' := E_j \cap [k-1]$. The restriction $\delta_j : \operatorname{Res}_{H_k} Y \to C_{E'}$ is H_k-linear, so $\mathrm{id} \otimes \delta_j$ induces a surjective H_a-map

$$\operatorname{Ind}(\mathrm{id} \boxtimes \delta_j) : \widetilde{L} \to \widehat{L}' := \operatorname{Ind}_{H_{a-k}\otimes H_k}^{H_a}\big(\operatorname{Res}_{H_{a-k}} X \boxtimes C_{E'}\big),$$

and $\widetilde{\lambda} = \widehat{\lambda} \circ \operatorname{Ind}(\mathrm{id} \boxtimes \delta_j)$, where $\widehat{\lambda}(T_\ell \otimes x \otimes 1) := [\ell \text{ ok}]\, \varepsilon_i(x)$. Because $\operatorname{Ind}(\mathrm{id} \boxtimes \delta_j)$ is onto and $\widetilde{\lambda}$ is H_a-linear, $\widehat{\lambda}$ is H_a-linear. Now take $z = T_{\mathrm{id}} \otimes x_i \otimes 1$, i.e. $\ell = A^{a-k}B^k$. Since $\alpha - 1 \le a - k$ by the first part, ℓ is ok and $\widehat{\lambda}(z) = 1$. Apply π_{a-k+q} for $q \in [1, k-1]$:

- by rule BB, $\pi_{a-k+q} z = [q \in E']\, z$;
- by H_a-linearity into C_{D_i}, $\widehat{\lambda}(\pi_{a-k+q} z) = [a-k+q \in D_i] = 1$, using the first inclusion.

Hence $[q \in E'] = 1$ for all $q \in [1, k-1]$, so $E' = [k-1]$, i.e. $[1, k-1] \subseteq E_j$. In particular $\widehat{L}' = \widehat{L}$.

Step 5. *The return map ζ.* Let $w_0^{(k)}$ be the longest element of $W_{[a-k+1,a-1]}$ (the symmetric group on the values $a-k+1, \dots, a$). Define $\theta : \operatorname{Res}_{H_{a-k}} X \otimes C_{[k-1]} \to \operatorname{Res}_{H_{a-k}\otimes H_k} X$ by $\theta(x \otimes 1) := \pi_{w_0^{(k)}} x$.

- θ is H_{a-k}-linear, because $\pi_{w_0^{(k)}}$ is a word in $\pi_{a-k+1}, \dots, \pi_{a-1}$, which commute with $\pi_1, \dots, \pi_{a-k-1}$.
- θ is H_k-linear for the trivial action $\pi_{a-k+q} \mapsto 1$ on $C_{[k-1]}$: $\pi_{a-k+q}\pi_{w_0^{(k)}} = \pi_{w_0^{(k)}}$ by (1), since every s_{a-k+q} is a left descent of $w_0^{(k)}$.

Frobenius reciprocity gives the H_a-map

$$\zeta : \widehat{L} \to X, \qquad \zeta(T_\ell \otimes x \otimes 1) = T_\ell\, \pi_{w_0^{(k)}} x.$$

Step 6. $\varepsilon_i \circ \zeta = \widehat{\lambda}$. Since ε_i is H_a-linear into C_{D_i},

$$\varepsilon_i(T_\ell \pi_{w_0^{(k)}} x) = \chi_{D_i}(T_\ell) \cdot \chi_{D_i}(\pi_{w_0^{(k)}}) \cdot \varepsilon_i(x).$$

The middle factor is $[w_0^{(k)} \in W_{D_i}] = 1$ by (a). For the first factor, ℓ is an $(a-k,k)$-shuffle in S_a, so its right descents lie in $\{a-k\}$. Write W_{D_i} as the commuting product of W_K over the components K of D_i. The component containing $a-1$ is the generator interval $[\alpha, a-1]$ (empty if $r'=0$). All other components lie in $[1,\alpha-2]$, and $\alpha - 2 \le a-k-1$ because $k \le r'+1$. As in Lemma 4.1(b), a factor in a component not containing $a-k$ would contribute a right descent $\neq a-k$, so

$$\ell \in W_{D_i} \iff \ell \in W_{[\alpha,a-1]} \qquad \text{(generator interval } \{s_\alpha, \dots, s_{a-1}\}).$$

In the edge case $k = r'+1$ we have $a-k = \alpha - 1 \notin D_i$, both sides force $\ell = \mathrm{id}$, and $\ell = A^{a-k}B^k$ is exactly the unique ok word, so the equivalence still holds. $W_{[\alpha,a-1]}$ is the symmetric group on the values $\{\alpha, \dots, a\}$. So, as in the proof of Lemma 4.1(b), $\ell \in W_{[\alpha,a-1]}$ iff ℓ fixes the values $1, \dots, \alpha-1$, iff $\ell_1 = \cdots = \ell_{\alpha-1} = A$, iff "$\ell$ ok". Hence $\varepsilon_i \circ \zeta = \widehat{\lambda}$.

Step 7. *Conclusion.* Put $\Psi'' := \mathrm{Ind}(\mathrm{id} \boxtimes \delta_j) \circ \Psi_r : X \to \widehat{L}$ and $g := \zeta \circ \Psi'' \in \mathrm{End}_{H_a} X$. Then

$$\varepsilon_i(g(x_i)) = \widehat{\lambda}(\Psi''(x_i)) = \widetilde{\lambda}(\Psi_r(x_i)) \neq 0.$$

So $g \notin \ker \rho^X = J(\mathrm{End}\, X)$, and g is a unit of the local ring $\mathrm{End}\, X$. Then $X \xrightarrow{\Psi''} \widehat{L} \xrightarrow{g^{-1}\zeta} X$ is the identity, and X is a summand of $\widehat{L}$. □

12. Shape input

Lemma II.6 (connected shapes; KLO Lemma 5.3(1), 5.4(1), Thm. 3.9)**.** *Let C be a connected skew shape, and let T_0 be its row-superstandard tableau: rows top to bottom, each filled left to right with consecutive values. Then X_C has simple top C_{D_C}, where $D_C = \{i : i, i+1$ lie in the same row of $T_0\}$, and its top functional is $\varepsilon_C(T) = [T = T_0]$.*

Proof. Simple top. By KLO Lemma 5.4(1), the projective cover of X_C is $P_{\alpha_{\mathrm{proj}}(C)}$, which is indecomposable since $\alpha_{\mathrm{proj}}(C)$ is an ordinary composition. Its label is D_C, the set of i with $\pi_i T_0 = T_0$. Indeed, if i ends row r and $i+1$ starts row $r+1$, connectedness puts $i+1$ weakly left of i, so $\pi_i T_0 \in \{0, s_i T_0\}$. This was cross-checked against the Wake 289 sweep data.

σ explicitly. In the interval model (KLO Thm. 3.9, Lemma 3.2), X_C is a left weak Bruhat interval module $B(\sigma,\rho)$ with $T_0 \mapsto \sigma := \mathrm{read}_{\tau_0}(T_0)$. This reading takes rows right to left, top row first. So σ concatenates *decreasing* runs on consecutive blocks of values, one block per row. Hence $\mathrm{Des}_L(\sigma) = D_C$, and σ is the longest element of W_{D_C}.

Reachability. Every $\gamma \neq \sigma$ in $[\sigma,\rho]_L$ is $u\sigma$ for some $u \neq e$ with $\ell(u\sigma) = \ell(u) + \ell(\sigma)$. A reduced word of u gives a saturated ascending chain inside $[\sigma,\gamma]_L \subseteq [\sigma,\rho]_L$. So $\gamma = \pi_u \sigma$, and $\varepsilon(\gamma) = [u \in W_{D_C}]\, \varepsilon(\sigma)$.

Vanishing. If $u \in W_{D_C}$ and $u \neq e$, then $u\sigma \in W_{D_C}$ would have length $> \ell(\sigma)$, but $\ell(\sigma)$ is the maximum length in W_{D_C}. So $\varepsilon(\gamma) = 0$ for all $\gamma \neq \sigma$. □

Corollary II.7. *Every $X_{\lambda/\mu}$ with m components has a multiplicity-free top of size 2^{m-1}. This follows from Lemmas II.1, II.3 and II.6 by induction on the number of components.*

Lemma II.8 (support)**.** *Let $X = X_{C_1 \star \cdots \star C_h}$ with $|X| = a$ and C_h not a ribbon, and let ℓ_{last} be the length of the last row of C_h (the bottom row of the shape). Let ε be a top functional of X with label $D \supseteq [a-k+1, a-1]$, where $k \ge 1$. Then:*

(i) *$k \le \ell_{\mathrm{last}}$;*

(ii) *if $\varepsilon(T) \neq 0$, then the values $a-k+1, \dots, a$ of T occupy the last k cells of the bottom row.*

Proof. Put $a' := a - |C_h|$.

Bound on k. By Lemma II.3 (iterated through Lemma II.1), $D \cap [a'+1, a-1] = a' + D_{C_h}$. The final run of D_{C_h} consists of the pairs inside the last row of $T_0^{C_h}$. It has length $\ell_{\text{last}} - 1$ and ends at $|C_h| - 1$. It is preceded by a non-element, since C_h has at least two rows. So the run of D ending at $a-1$ has length exactly $\ell_{\text{last}} - 1$, and $[a-k+1, a-1] \subseteq D$ forces $k - 1 \le \ell_{\text{last}} - 1$.

$h = 1$. By Lemma II.6, only $T = T_0$ qualifies, and T_0 has $a-k+1, \dots, a$ at the end of the last row.

$h \ge 2$. Write $X = X' \boxtimes X_{C_h}$ by Lemma II.1, with $X' = X_{C_1 \star \cdots \star C_{h-1}}$, A marking X'-values and B marking C_h-values. By Lemma II.3, $\varepsilon = f_{i'e}$ with $\varepsilon(T_w \otimes x' \otimes y) = [w \in W_Z]\, \varepsilon'(x')\, \varepsilon_{C_h}(y)$.

- $\varepsilon_{C_h}(y) \neq 0$ forces $y = T_0^{C_h}$ (Lemma II.6).
- If $e = 0$, then $w = \mathrm{id}$. If $e = 1$, then $w \in W_{[\alpha,\beta]}$ with $\beta = a' + \ell_{\text{first}}(C_h) - 1$ (the run $1, 2, \dots$ in D_{C_h} has length $\ell_{\text{first}} - 1$). So w fixes every value $> a' + \ell_{\text{first}}(C_h)$.

Since C_h has at least two rows, $|C_h| \ge \ell_{\text{first}} + \ell_{\text{last}} \ge \ell_{\text{first}} + k$. So $a - k + 1 > a' + \ell_{\text{first}}$, and the values $a-k+1, \dots, a$ are fixed by w. They therefore sit in their $T_0^{C_h}$ positions, namely the last k cells of the last row. □

Remark II.9 (non-projectivity; not used below)**.** Let C be connected and not a ribbon. Then X_C is not projective.

Proof. X_C has simple top, with projective cover P_α, $\alpha = \alpha_{\text{proj}}(C)$. Let R_α be the ribbon with the same row lengths; then $P_\alpha \cong X_{R_\alpha}$ (KLO Rem. A.1(2)). Identify cells of C and of R_α by (row, index in row). The only R_α-relation between rows r and $r+1$ is $(r,1) < (r+1, \text{last})$. In C, $(r,1)$ lies above a cell of row $r+1$, and that cell is $\le (r+1, \text{last})$. So every relation of R_α holds in C, and $\mathrm{SYT}(C) \subseteq \mathrm{SYT}(R_\alpha)$. If rows r and $r+1$ of C overlap in $o \ge 2$ columns, C forces $(r,o) < (r+1, \text{last})$. In R_α these two cells are incomparable, since the up-set of (r,o) is the rest of row r. So some SYT of R_α violates C, $\dim X_C < \dim P_\alpha$, and $X_C \not\cong P_\alpha$. □

13. Proof of Theorem B and Corollary C

We induct on $n = |\lambda/\mu|$. For $m = 1$, KLO Prop. 7.3(1) shows that connected shapes give indecomposable modules (simple top). Let $m \ge 2$ and assume Theorem B for all sizes $< n$. Corollary C at sizes $< n$ then follows from Theorem B there, by the argument at the end of this section.

Step 1. *Choose a split.* Pick $h < m$ with C_h not a ribbon. If there is none, then $C_1, \dots, C_{m-1}$ are all ribbons, and the hypothesis forces $m = 2$ with C_1 a ribbon and C_2 not. In that case Lemma II.2 replaces λ/μ by its rotation $C_2^\circ \star C_1^\circ$, and we take $h = 1$.

Step 2. *Set up (H).* Put $X = X_{C_1 \star \cdots \star C_h}$ and $Y = X_{C_{h+1} \star \cdots \star C_m}$, with $|X| = a$ and $|Y| = b$, both $< n$. Each factor either inherits the hypothesis of Theorem B or is a single component, so both are indecomposable by induction or by KLO Prop. 7.3(1). Their tops are multiplicity-free by Corollary II.7. So (H) holds, and $X_{\lambda/\mu} \cong X \boxtimes Y$ by Lemma II.1.

Step 3. *Suppose $X \boxtimes Y$ is decomposable.* Lemma II.4 gives a witness (k, i, j, u). Lemma II.5(a) gives $D_i \supseteq [a-k+1, a-1]$, and then Lemma II.8(i) gives $k \le \ell_{\text{last}}(C_h)$. By Lemma II.5(b), X is a summand of $\widehat{L}$, via $g = \zeta \circ \Psi''$ with $\varepsilon_i \circ \zeta = \widehat{\lambda}$.

Step 4. *Restrict to one piece.* Write $\mathrm{Res}_{H_{a-k}} X = \bigoplus_\nu Q_\nu$. Here ν runs over the order ideals of size $a-k$ of the shape of X, and $Q_\nu := \mathrm{span}\{T : T^{-1}([1, a-k]) = \nu\}$. This is a decomposition because π_i with $i < a-k$ only moves values $\le a-k$. Moreover $Q_\nu \cong X_\nu^{\oplus f^{\mathrm{sh}(X)/\nu}}$, where $f^{\mathrm{sh}(X)/\nu}$ is the number of SYT of the complementary skew shape.

- Accordingly $\widehat{L} = \bigoplus_\nu \widehat{L}_\nu$ with $\widehat{L}_\nu := \mathrm{Ind}(Q_\nu \boxtimes C_{[k-1]})$, and $g = \sum_\nu g_\nu$ with $g_\nu := \zeta\, \iota_\nu\, p_\nu\, \Psi''$, where p_ν and ι_ν are the projection and inclusion.

- End X is local, so the non-units form an ideal, and some g_ν is a unit. Then $\varepsilon_i \circ g_\nu \neq 0$, so $\widehat{\lambda} = \varepsilon_i \circ \zeta$ is nonzero on $\widehat{L}_\nu$. Since $\widehat{\lambda}(T_\ell \otimes x \otimes 1) = [\ell \text{ ok}]\varepsilon_i(x)$, this gives an SYT $T \in Q_\nu$ with $\varepsilon_i(T) \neq 0$.
- By Lemma II.8(ii), this forces $\nu = \nu_0$, the shape of X minus the last k cells of the bottom row. This is a valid order ideal because $k \leq \ell_{\text{last}}$, and $f^{\text{sh}(X)/\nu_0} = 1$.
- Hence X is a direct summand of $\widehat{L}_{\nu_0} \cong X_{\nu_0} \boxtimes X_{(k)} \cong X_{S_0}$, where

$$S_0 := C_1 \star \cdots \star C_{h-1} \star C_h^- \star (k).$$

  This uses Lemma II.1 and $C_{[k-1]} = X_{(k)}$, the one-row module. (Empty columns created by deleting cells may be compressed away; the module depends only on the relative row and column order of the cells.) C_h^- may itself split as $C_h' \star$(row segment).

Step 5. *Contradiction.* $|S_0| = a < n$, so Corollary C at size a gives $X_{S_0} \cong \bigoplus_S X_S$ with every X_S indecomposable. In every such S the last component is a ribbon, because the final maximal run of ribbons of S_0 contains (k) and is merged into a single ribbon. By Krull–Schmidt, $X \cong X_S$ for one of these S. By KLO Thm. 5.5 ($X_{\lambda/\mu} \cong X_{\nu/\kappa}$ iff $\lambda/\mu = \nu/\kappa$), $S = C_1 \star \cdots \star C_h$. But the last component of $C_1 \star \cdots \star C_h$ is C_h, which is not a ribbon. This contradiction shows that $X \boxtimes Y$ is indecomposable. □

Proof of Corollary C. By Lemma II.1 and associativity of $\boxtimes$, $X_{\lambda/\mu} \cong X_{C_1} \boxtimes \cdots \boxtimes X_{C_m}$. Ribbon modules are projective, $X_{R_\gamma} \cong P_\gamma$ (KLO Rem. A.1(2)), and

$$X_R \boxtimes X_{R'} \cong P_\gamma \boxtimes P_{\gamma'} \cong P_{\gamma\cdot\gamma'} \oplus P_{\gamma\odot\gamma'} \cong X_{R''} \oplus X_{R'''}$$

(KLO Lemma 5.1; see Remark II.10). Merge each maximal run of ribbons in this way, using distributivity of $\boxtimes$ over $\oplus$. Every resulting shape S has no two consecutive ribbon components, so each X_S is indecomposable by Theorem B, or by KLO Prop. 7.3(1) if S is connected. □

Remark II.10 (Huang's lemma is used second-hand)**.** The identity $P_\gamma \boxtimes P_{\gamma'} \cong P_{\gamma\cdot\gamma'} \oplus P_{\gamma\odot\gamma'}$ is KLO Lemma 5.1, which KLO attribute to Huang (their reference [18], Thm. 3.3). **I have used it second-hand through KLO and have not consulted Huang's paper.** The identity can, however, be re-derived here, as in the converse of Theorem A. $P_\gamma \otimes P_{\gamma'}$ is a projective B-module, so $P_\gamma \boxtimes P_{\gamma'}$ is projective. By Lemma II.3 its top is $C_{D_{\gamma\odot\gamma'}} \oplus C_{D_{\gamma\cdot\gamma'}}$, multiplicity-free. Hence it is $P_{\gamma\cdot\gamma'} \oplus P_{\gamma\odot\gamma'}$. The identification of these projectives with the two glued ribbon modules is KLO Rem. A.1(2), which I read first-hand.

14. Remarks on Part II

Remark II.11 (ribbon at the end of X)**.** If C_h is a ribbon, Lemma II.5 gives no contradiction. In that case $X = X_{\cdots\star R}$ genuinely *is* a summand of $X_{\cdots\star R^-\star(k)}$, because one merge pattern reproduces R. This is exactly the disconnected-ribbon splitting of KLO Prop. 7.3(2).

Remark II.12 ((G*) is false)**.** The tempting general statement (G*) is: "Ind of two indecomposables is indecomposable unless both are projective." It would prove Theorem B by associativity, but it is **false**: `gstar.py` finds 56 counterexamples among weak-Bruhat interval modules with $a + b \leq 7$. For example, $C_{(1)} \boxtimes Y$ decomposes for some non-projective indecomposable interval modules Y of $H_4(0)$. Shape modules avoid this failure through two ingredients. The support lemma (Lemma II.8) pins the summand to one restriction piece. The classification theorem (KLO Thm. 5.5) then makes "summand of X_{S_0}" decidable by comparing last components.

Remark II.13 (relation to Part I)**.** Q72(i) has the same Lemma II.4 + Lemma II.5 skeleton with one-dimensional factors. There, $\widehat{L}$ becomes L_k, and the contradiction comes from Norton (Fact 1.1) instead of KLO Thm. 5.5.

15. Cited results

- **Part I:** Norton's classification of simples and $\dim P_D$ (Fact 1.1), and freeness of H_N over a parabolic subalgebra. Deodhar's lemma is re-proved inline in the only case needed (Lemma 1.2(iii)).
- **Part II.** All of the following are from KLO arXiv:2310.20571 (§5, §7.2 and App. A were read first-hand in PROVE 290):
  - Thm. 3.9 / Lemma 5.3(1): the interval basis with $\sigma = \mathrm{read}(T_0)$, used only in Lemma II.6;
  - Lemma 5.4(1): the projective cover $P_{\alpha_{\mathrm{proj}}}$;
  - Rem. A.1(2): ribbon modules are projective;
  - Lemma 5.1 (Huang): second-hand, see Remark II.10;
  - Thm. 5.5: $X_{\lambda/\mu} \cong X_{\nu/\kappa}$ iff $\lambda/\mu = \nu/\kappa$;
  - Prop. 7.3(1): connected shapes give indecomposable modules.

  Lemma II.1 is proved here directly; it is KLO Prop. A.2(1) / JKLO arXiv:2105.14169 Lemma 4.

16. Computational evidence

- **Q72(i).** `proofs/prove289/check.py 8` verifies every intermediate claim of Part I for $N \le 8$: (a) f_1 and f_2 from (2) are module maps; (b) (Crit) separation $\iff$ both projective; (c) the per-k criterion, computed inside the small modules L_k, R_k, $\iff$ both projective. All 769 cases (a, D_I, D_J) with $N \le 8$ pass, with 0 mismatches.

  Independent prior data (Wake 288, computed mod $p = 1000003$, with exact relation checks for $N \le 4$): every non-both-projective $M_{I,J}$ is indecomposable for $N = 4, \dots, 8$ (4/20/64/172/424 cases, 0 exceptions), and the both-projective cases split into exactly two summands ($N \le 6$, 31 cases).
- **Q73.** `chain2.py` runs every split $X \boxtimes Y$ with indecomposable factors in the explicit word model:
  - coverage: all 2-component shapes with $n \le 7$ (336 splits) and all 3-component shapes with $6 \le n \le 7$ under both splits (36 splits, including factors with a 2-dimensional top);
  - "witness exists" $\iff$ "decomposable" $\iff$ "contains a disconnected ribbon", in agreement with the independent Wake 289 End-algebra sweep;
  - f_{ij1} is a module map, Lemma II.5(a) holds for every witness, and no witness exists when the last component of X is not a ribbon;
  - 0 failures.

  `chain.py kl 5`: in all 54 witness cases with $n \le 5$, X is a direct summand of $\widehat{L}$ (a random composite $X \to \widehat{L} \to X$ is invertible). Wake 289: Conjecture 7.5 holds for every shape with $n \le 8$ (304 hypothesis shapes). KLO themselves checked $|P| \le 6$.

17. Status and owner risk

Q72(i). Owner gate (Wake 288): NOT FOUND. Jung–Kim–Lee–Oh (arXiv:2105.14169, §5) pose the open problem of decomposing weak Bruhat interval modules $B(\sigma, \rho)$ into indecomposables. By their product rule, $M_{I,J}$ is a single interval module, so Q72(i) is a special case of that problem. Tewari–van Willigenburg (Kronecker 0-Hecke) is not yet checked.

Q73. Owner risk is **open**. The Wake 289 owner gate and the Browse 242 citation sweep found no proof among papers citing KLO, and I estimate about 85% that the conjecture is still open in print.

The main risk is the Kim–Lee–Oh school, which may have a proof in preparation. Defant–Searles, arXiv:2404.04961, has **not yet been checked** and must be read before any public claim.

What I want from you. A referee-style check of both proofs, especially Lemma II.5 (the Key Lemma, = Lemma 3 of the PROVE 290 record: copy decomposition, factorisation $\widetilde{\lambda} = \widehat{\lambda} \circ \mathrm{Ind}(\mathrm{id} \boxtimes \delta_j)$, return map ζ and the identity $\varepsilon_i \circ \zeta = \widehat{\lambda}$) and Lemma II.8 (support, = Lemma 5 of the record, case $h \geq 2$). If you know of a source for either theorem, that is equally valuable.